

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

10 answers · Rationale-rich · Teach from doc

25

Q1**D****D.** 84

Choice D is correct. The value of fx when $x = -8$ can be found by substituting -8 for x in the given function. This yields $f(-8) = -3(-8) + 60$, or $f(-8) = 84$. Therefore, when $x = -8$, the value of fx is 84.

Choice A is incorrect. This is the value of $-3 + -8 + 60$, not $-3(-8) + 60$.

Choice B is incorrect. This is the value of $-8 + 60$, not $-3(-8) + 60$.

Choice C is incorrect. This is the value of $-3 + 60$, not $-3(-8) + 60$.

Q2**C****C.** $fx = 17$

Choice C is correct. An equation defining the linear function f can be written in the form $fx = mx + b$, where m is the slope and b is the y -intercept of the graph of $y = fx$ in the xy -plane. The slope of the graph of $y = fx$ can be found using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$, where x_1, y_1 and x_2, y_2 are any two points that the graph passes through. If $f(0) = 17$, it follows that the graph of $y = fx$ passes through the point $(0, 17)$. If $f(1) = 17$, it follows that the graph of $y = fx$ passes through the point $(1, 17)$. Substituting $(0, 17)$ and $(1, 17)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{17 - 17}{1 - 0}$, which is equivalent to $m = \frac{0}{1}$, or $m = 0$. Since the graph of $y = fx$ passes through $(0, 17)$, it follows that $b = 17$. Substituting 0 for m and 17 for b in the equation $fx = mx + b$ yields $fx = 0x + 17$, or $fx = 17$. Thus, the equation that defines f is $fx = 17$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**C**

C. The object is moving at a rate of 16 inches per second.

Choice C is correct. It's given that in the equation $d = 16t$, d represents the distance, in inches, and t represents the number of seconds since an object started moving. In this equation, t is being multiplied by 16. This means that the object's distance increases by 16 inches each second. Therefore, the best interpretation of 16 in this context is that the object is moving at a rate of 16 inches per second.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This is the best interpretation of $16t$, rather than 16, in this context.

Choice D is incorrect and may result from conceptual errors.

Q4**D**

D. 8

Choice D is correct. The value of $g(0)$ is found by substituting 0 for x in the function g . This yields $g(0) = -0 + 8$, which can be rewritten as $g(0) = 8$.

Choice A is incorrect and may result from misinterpreting the equation as $g(x) = x + (-8)$ instead of $g(x) = -x + 8$. Choice B is incorrect. This is the value of x , not $g(x)$. Choice C is incorrect and may result from calculation errors.

Q5**B****B.** \$ 4.25

Choice B is correct. It's given that Josie purchased a pass to tour a nature preserve and 10 identical stickers for \$ 62.00 total. It's also given that the pass cost \$ 19.50. Let x represent the cost, in dollars, of one sticker. This relationship can be represented by the equation $19.50 + 10x = 62$. Subtracting 19.50 from both sides of this equation yields $10x = 42.50$. Dividing both sides of this equation by 10 yields $x = 4.25$. Therefore, the cost of one sticker was \$ 4.25.

Choice A is incorrect. This is the cost of one sticker if Josie purchased 10 identical stickers for \$ 19.50, not a pass and 10 identical stickers for \$ 62.00.

Choice C is incorrect. This is the cost of one sticker if Josie purchased 10 identical stickers for \$ 62.00, not a pass and 10 identical stickers for \$ 62.00.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**C****C.** 175

Choice C is correct. The line shown represents the cost y , in dollars, for a manufacturer to make x rings. For the line shown, the x -axis represents the number of rings made by the manufacturer and the y -axis represents the cost, in dollars. Therefore, the cost, in dollars, for the manufacturer to make 60 rings is represented by the y -coordinate of the point on the line that has an x -coordinate of 60. The point on the line with an x -coordinate of 60 has a y -coordinate of 175. Therefore, the cost, in dollars, for the manufacturer to make 60 rings is 175.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the cost, in dollars, for the manufacturer to make 20 rings.

Choice D is incorrect. This is the cost, in dollars, for the manufacturer to make 100 rings.

Q7**A****A.** 10

Choice A is correct. It's given that the function f is defined by $fx = 5x + 8$. Substituting 58 for fx in this equation yields $58 = 5x + 8$. Subtracting 8 from both sides of this equation yields $50 = 5x$. Dividing both sides of this equation by 5 yields $10 = x$. Therefore, the value of x when $fx = 58$ is 10.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of fx , not the value of x when $fx = 58$.

Q8**D****D.** The estimated height of the tree was 4 feet when it was first measured.

Choice D is correct. It's given that the function $fx = 8x + 4$ gives the estimated height, in feet, of a willow tree x years after its height was first measured. For a function defined by an equation of the form $fx = mx + b$, where m and b are constants, b represents the value of fx when $x = 0$. It follows that in the given function, 4 represents the value of fx when $x = 0$. Therefore, the best interpretation of 4 in this context is that the estimated height of the tree was 4 feet when it was first measured.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q9**A**

A. $h = 8s + 15$

Choice A is correct. It's given that the front of the roller-coaster car starts rising when it's 15 feet above the ground. This initial height of 15 feet can be represented by a constant term, 15, in an equation. Each second, the front of the roller-coaster car rises 8 feet, which can be represented by $8s$. Thus, the equation $h = 8s + 15$ gives the height, in feet, of the front of the roller-coaster car s seconds after it starts up the hill.

Choices B and C are incorrect and may result from conceptual errors in creating a linear equation. Choice D is incorrect and may result from switching the rate at which the roller-coaster car rises with its initial height.

Q10**A**

A. \$46.60

Choice A is correct. It's given that Robert was charged an initial fee of \$20.00 to rent the truck plus an additional \$0.70 per mile driven. Let m represent the number of miles the truck was driven. Since the rental charge is \$0.70 per mile driven, $0.70m$ represents the amount Robert was charged for m miles driven. Let c equal the total amount, in dollars, Robert was charged to rent the truck. The total amount can be represented by the equation $c = 20.00 + 0.70m$. It's given that the truck was driven 38 miles, thus $m = 38$. Substituting 38 into the equation gives $c = 20.00 + 0.70(38)$. Multiplying $0.70(38)$ gives $c = 20.00 + 26.60$. Adding these values gives $c = 46.60$, so the total amount Robert was charged is \$46.60.

Choices B, C, and D are incorrect and may result from setting up the equation incorrectly or from making calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

